Technical Report

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Project: Sleep Apnea Detection Using Physiological Signals and Deep Learning

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1. Introduction

Sleep apnea is a condition in which breathing repeatedly stops and starts during sleep. It affects an estimated 1 billion people worldwide and remains chronically underdiagnosed due to the high cost and limited availability of clinical sleep studies. Automated detection of breathing irregularities from physiological signals can significantly reduce the burden of manual annotation and enable large scale screening.

This project builds an end to end machine learning pipeline that takes raw overnight polysomnography recordings as input and outputs a trained classifier capable of detecting Normal breathing, Hypopnea, and Obstructive Apnea events from 30-second signal windows.

2. Dataset

The dataset consists of overnight polysomnography recordings from 5 participants (AP01 to AP05). Each participant folder contains three physiological signals recorded simultaneously during sleep:

Signal	Description	Sampling Rate	Samples per 8 Hours
Nasal Airflow	Measures airflow through the nose	32 Hz	921,600
Thoracic Movement	Measures chest and abdominal expansion	32 Hz	921,600
SpO2	Blood oxygen saturation level	4 Hz	115,200

Each folder also contains a flow events file with expert annotated breathing events including start time, end time, and event type. A sleep profile file provides sleep stage annotations every 30 seconds covering Wake, N1, N2, N3, and REM stages.

3. Signal Preprocessing

3.1 Bandpass Filtering

Human breathing typically occurs at a rate of 10 to 24 breaths per minute, corresponding to a frequency range of 0.17 Hz to 0.4 Hz. Raw physiological signals contain noise from body movement, electrical interference, and sensor artifacts at frequencies outside this range. A 4th order Butterworth bandpass filter was applied to all three signals using scipy's butter and filtfilt functions. filtfilt applies the filter

forward and backward, eliminating phase distortion.

3.2 Windowing

The filtered signals were segmented into 30-second windows with 50% overlap. This window length captures multiple complete breathing cycles while remaining short enough to localize specific events.

Signal	Sampling Rate	Window Size	Step Size
Nasal Airflow	32 Hz	960 samples	480 samples
Thoracic Movement	32 Hz	960 samples	480 samples
SpO2	4 Hz	120 samples	60 samples

3.3 Labeling Strategy

Each window was labeled using the flow events annotation file. If a labeled breathing event overlapped with more than 50% of the window duration, that event label was assigned to the window. If no event satisfied this threshold, the window was labeled Normal. A second dataset was also created using the sleep profile file where each window was labeled with the sleep stage corresponding to the closest timestamp.

4. Models

4.1 1D Convolutional Neural Network

A 1D CNN was designed to classify each 30-second window. The input shape is (batch, 3, 960) where the 3 channels correspond to nasal airflow, thoracic movement, and SpO2 upsampled from 120 to 960. The architecture consists of three convolutional blocks with increasing filter sizes (32, 64, 128), each followed by batch normalization, ReLU activation, and max pooling. Global average pooling collapses the time dimension into a 128-dimensional feature vector, passed through fully connected layers with dropout (0.5) for final classification.

4.2 Conv-LSTM

A Conv-LSTM model was also implemented to capture both local patterns and sequential dependencies. The same convolutional frontend extracts features which are reshaped and fed into a 2-layer LSTM with hidden size 64. The final hidden state is passed to the classifier, benefiting from the LSTM's ability to remember patterns across the time dimension.

4.3 Training Details

Both models were trained using CrossEntropyLoss and the Adam optimizer with a learning rate of 0.001 for 30 epochs. Features were normalized using StandardScaler fitted on training data only to prevent data leakage. Batch size was set to 32.

5. Evaluation Strategy

Leave-One-Participant-Out (LOPO) cross validation was used to evaluate model generalization. In each of the 5 folds, one participant was held out as the test set while the model was trained on the remaining four. This strategy simulates real world deployment where the model must generalize to a new patient it has never seen during training.

6. Results

6.1 Per-Fold Results

Participant	Accuracy	Precision	Recall
AP01	0.9479	0.8984	0.9479
AP02	0.9135	0.8345	0.9135
AP03	0.9900	0.9801	0.9900
AP04	0.9125	0.8327	0.9125
AP05	0.7944	0.6311	0.7944
Overall	**0.9137**	**0.8349**	**0.9137**

6.2 Confusion Matrix

The confusion matrix below shows predictions across all 5 folds combined. Classes: Body event, Hypopnea, Mixed Apnea, Normal, Obstructive Apnea.

True \ Pred	Body event	Hypopnea	Mixed Apnea	Normal	Obst. Apnea
Body event	0	0	0	3	0
Hypopnea	0	0	0	593	0
Mixed Apnea	0	0	0	2	0
Normal	0	0	0	8041	0
Obst. Apnea	0	0	0	161	0

6.3 Discussion

The model achieved an overall accuracy of 91.37% across all 5 LOPO folds. AP03 showed the strongest performance with 99% accuracy while AP05 was the most challenging participant with 79.44% accuracy. This variance reflects natural differences in individual physiology and is expected in LOPO evaluation.

The confusion matrix reveals the model predominantly predicts the Normal class — a known challenge with imbalanced datasets. Future work should explore class weighting, SMOTE oversampling, or threshold tuning to improve recall on minority classes. In a clinical context higher recall on apnea classes would be preferred to avoid missed detections.

7. Conclusion

This project demonstrates an end to end pipeline for automated sleep apnea detection from polysomnography signals, covering signal visualization, bandpass filtering, windowing, labeling, and classification using a 1D CNN with LOPO cross validation. The model achieves 91.37% overall accuracy across 5 unseen participants. Key areas for improvement include addressing class imbalance and extending evaluation to the Conv-LSTM model.

AI Tool Disclosure: Deep learning components were developed with assistance from Claude (Anthropic). Signal preprocessing, visualization, and pipeline design were implemented independently by the author.